Assignment4
Part 1

A4 Requirements:

1. cd to local **repos** subdirectory:
   Mac: `cd /Applications/Ampps/www/repos`
   Windows: `cd "c:\Program Files(x86)\Ampps\www\repos"`
   Note: *Be careful*! Windows **must** use quotation marks because of spaces in directory names. ☹
   Note: If you do *not* have AMPPS cd to *your* **www** directory.

2. Clone assignment starter files:
   `git clone https://bitbucket.org/mjowett/lis4381_student_files`
   (clones **lis4381_student_files** Bitbucket repo)

3. Review subdirectories and files

4. Open **index.php** and review code:
   a. Suitably modify meta tags
   b. Change title, navigation links, and header tags appropriately
   c. Add form controls to match attributes of **petstore** entity
   d. Add the following jQuery validation and regular expressions-- as per the entity attribute requirements (and screenshots below):
      i. ***All*** input fields, except Notes **are required**
      ii. Use min/max jQuery validation
      iii. Use **regexp** to only allow appropriate characters for each control:
      pst_name, pst_street, pst_city, pst_state, pst_zip, pst_phone, pst_email, pst_url, pst_ytd_sales, pst_notes
      Name (provided)**:** letters, numbers, hyphens, and underscore
      Street, City (provided)**:**
      no more than 30 characters
      Street: must only contain letters, numbers, commas, hyphens, or periods
      City: can only contain letters, numbers, hyphens, and space character (29 Palms)
      State:
      must be 2 characters
      must only contain letters
      Zip:
      must be between 5 and 9 digits, inclusive
      must only contain numbers
      Phone:
      must be 10 digits, including area code
      must only contain numbers
      Email: See http://www.qcitr.com/usefullinks.htm#lesson7
      URL:
      no more than 100 characters
      See http://www.qcitr.com/usefullinks.htm#lesson7
      YTD Sales:
      no more than 10 digits, including decimal point
      can only contain numbers, and decimal point (if used)
      iv. ***After*** testing jQuery validation, use HTML5 property to limit the number of characters for each control
      v. Research what the following validation code does:

```
valid: 'fa fa-check',
invalid: 'fa fa-times',
validating: 'fa fa-refresh'
```

5. Create a **favicon** using ***your*** initials, and place it in each assignment's main directory, including **lis4381**: Example: https://www.favicon-generator.org/

Developer Notes:

- A1, A2, A3, and P1 links ***must*** work, ***and*** each index.php file should have content, as per the associated Bitbucket README.md files.
- Research: bootstrap responsive image size, including images w/in the carousel component.

Helper Videos:

1. http://www.qcitr.com/vids/LIS4381_A4a.mp4
2. http://www.qcitr.com/vids/LIS4381_A4b.mp4

Note:

Be sure to review the include files to see what they do, and how they work--though, **do *not* change them**! After the end of the semester, you can modify these files to test newer versions of Bootstrap--**but, wait until all course work has been completed**.

Part 2

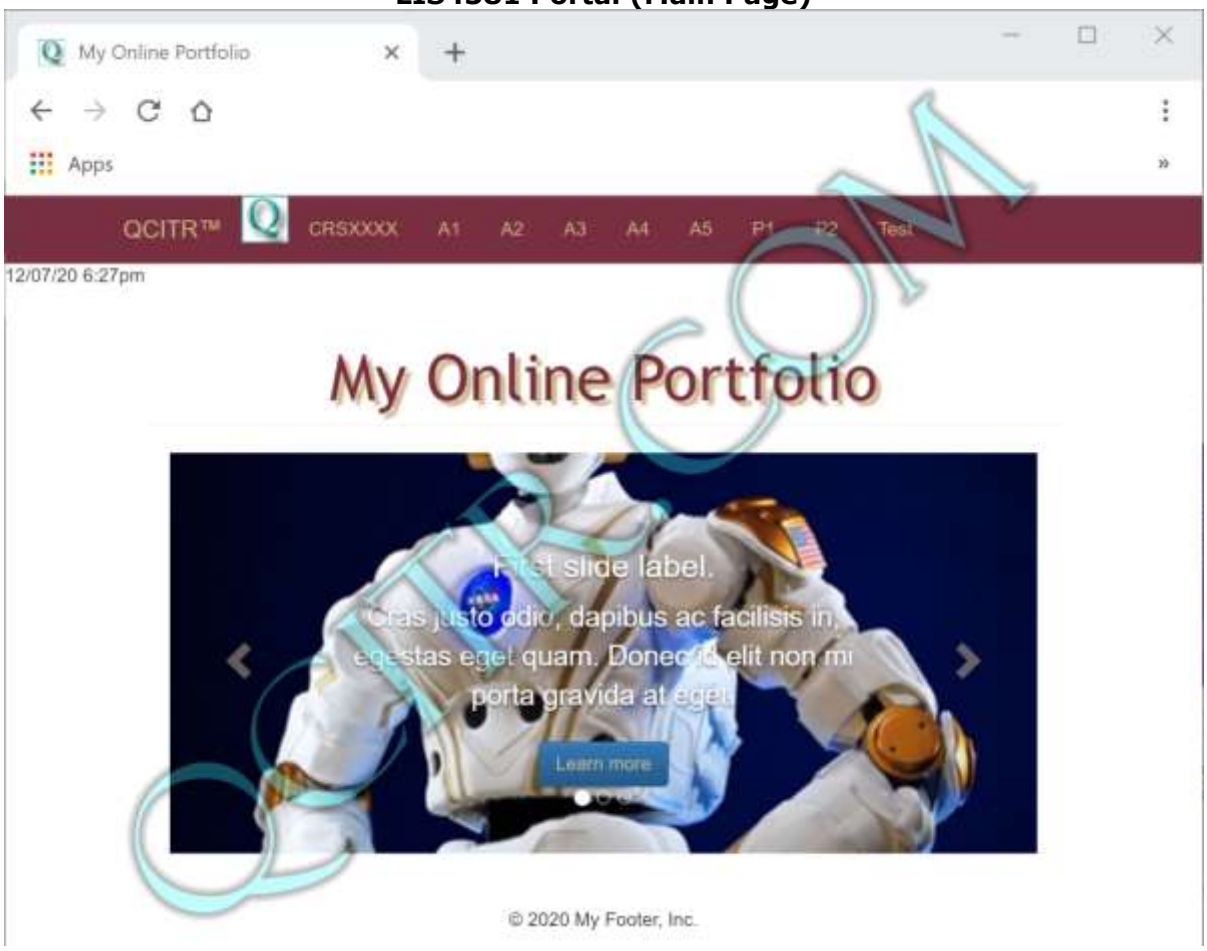
README.md file should include the following items:

1. **Course title, your name, assignment requirements, as per A1;**
2. Screenshots as per below examples;
3. Link to **local lis4381 web app**: http://localhost/repos/lis4381/

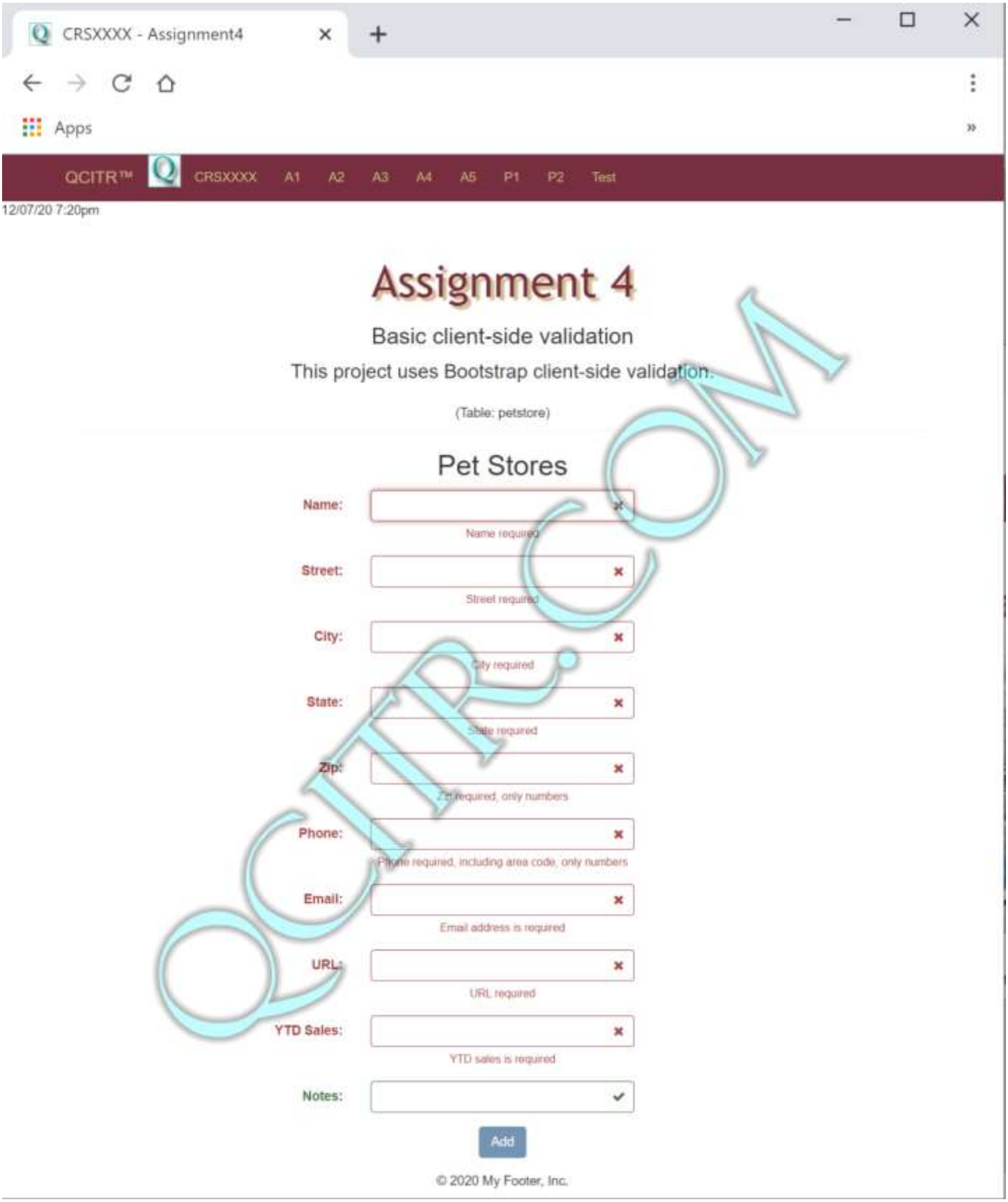
Deliverables (see screenshots below):

1. ***MUST*** provide **Bitbucket** read-only access to **course** repo.
2. **README.md** must include screenshots per above (see examples below).
3. **FSU's Learning Management System**: **Bitbucket** repo link

LIS4381 Portal (Main Page)



Failed Validation



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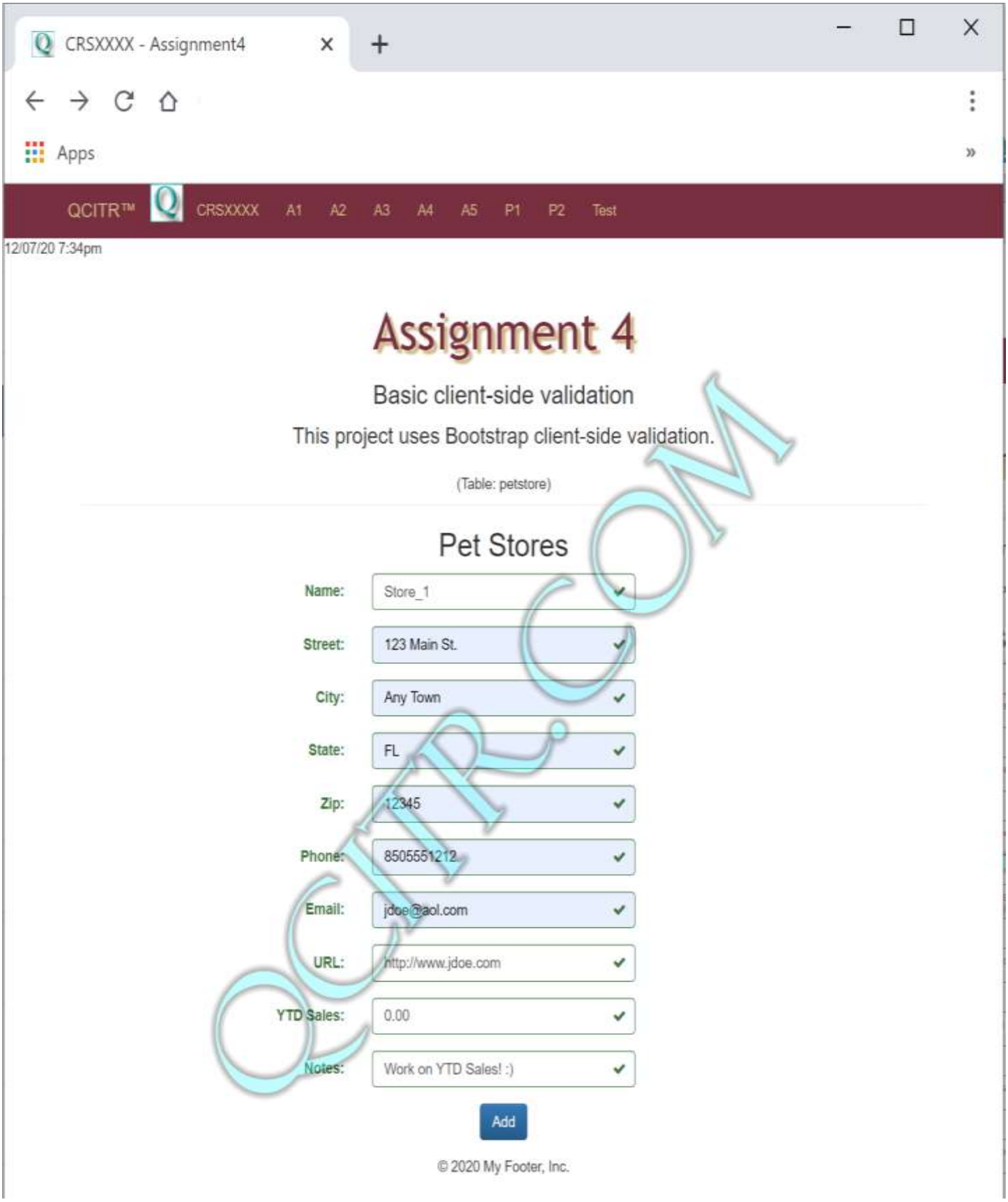
Passed Validation



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Part 3
Questions (PHP/MySQL: Chs. 9, 10, 15):

1. Which of the following functions returns the length of a string (i.e., number of characters that it contains)?

a. stringlength() b. strlen() c. length() d. strlength()

2. If you want to round a number within an arithmetic expression, which function should you use?

a. number_round b. round c. sprintf d. round_num

3. Using strnatcasecmp($str1, $str2) function for a "natural" comparison of two values

a. is case-sensitive
b. returns -1 if $str2 comes before $str1
c. would evaluate both "09" and "9" as being before "10"
d. would evaluate "10" as being before both "09" and "9"

4. Why is it important not to begin an integer value with a zero (e.g., 03), when working with numbers in calculations?

a. Because it would be treated as a hexadecimal value, and yield an incorrect result.
b. Because it would be treated as a decimal value, and yield an incorrect result. c. Because it would be treated as an octal value, and yield an incorrect result.
d. Because it would be treated as a whole number, and yield an incorrect result.

5. When you create a DateInterval object, you can use the format() method, and pass it a number of arguments that specifies a format code for one or more of the following descriptions:

a. years, months, days
b. hours, minutes, seconds
c. years, months, days, hours, minutes, seconds
d. years, months, days, hours, minutes, seconds, and sign (+ or -) of the date interval

6. When you use the substr function to extract a substring from a string, you need to at least pass an argument that gives

a. the characters in the substring that you're looking for
b. the starting position of the substring
c. the length of the substring
d. the separators for the substring

```
$current_date = new DateTime();
$due_days_diff = $current_date->diff($due_date);
if ($current_date > $due_date) {
    $overdue_message = $due_days_diff->format(
        '%y years, %m months, and %d days overdue.');
}
```

7. If $due_date contains a DateTime object, $due_date_diff will contain

a. a TimeStamp object
b. a DateTime object
c. a DateInterval object
d. a TimeInterval object

8. In order to print out a variable's value, when you code a variable name within a string literal, you have to enclose the literal within _____ quotation marks.

a. either double or single
b. single
c. double
d. neither double or single

9. A DateInterval object represents a/an _____ time.

a. period of
b. point in
c. exact
d. instance in

10. A substring within a string is identified by its _____ and its _____ within the string.

a. length, ending position
b. length, starting position
c. value, starting position
d. length, value

11. A timestamp stores the number of _____ since midnight on January 1, 1970 GMT.

a. Milliseconds
b. Hours
c. Minutes
d. seconds

12. After you create a DateTime object, you can use its setTime method to change the time or its _____ method to change the date.

a. setDateValue
b. setValueDate
c. setDate
d. setValue

13. One of the methods of a _____ object lets you add a DateInterval object to the object.

a. TimeInstance
b. Date
c. DateInstance
d. DateTime

14. Because most systems today store timestamps as 32-bit signed integers, the upper limit of a timestamp on these systems is January 19 in the year _____.

a. there is no year limit
b. 2099
c. 2038
d. 2199

15. By using _____, you can put special characters into a string literal that would otherwise be interpreted incorrectly by PHP.

a. escape sequences
b. n2br function
c. ASCII characters
d. ANSI characters

16. By using the _____ function, you can send special characters to the browser in a way that they will be displayed correctly.

a. xmlentities
b. html5entities
c. htmlentities
d. xhtmlentities

17. The _____ function returns the largest value of the two or more arguments that are passed to it.

a. ceil()
b. max()
c. largest()
d. maximum()

18. If a number in a PHP application is larger than the maximum value that PHP provides for, it is converted into the PHP constant named _____.

a. OUT_OF_BOUNDS
b. MAXIMUM
c. INFINITE
d. INF

19. The date function makes it easy to create a timestamp for the current date and then _____ before returning the value.

a. convert it to Unix Epoch time
b. convert it to timestamp
c. return a string formatted as a date
d. convert it to numbers

20. The diff method of a DateTime object requires another DateTime object as its argument and returns a _____ object.

a. DateInterval
b. IntervalDate
c. DateInt
d. Date

21. The explode function returns an array of strings from a single string based on the _____ character that you specify.

a. HTML
b. XML
c. delimiter
d. ISO

22. The ltrim and rtrim functions let you _____ from the front or back of a string.

a. replace all spaces
b. remove all spaces
c. remove all characters
d. include all characters

23. The strcasecmp function is _____. As a result, “Anders” comes before “zylka”.

a. case-corrective
b. case-specific
c. case-sensitive
d. case-insensitive

24. The use of _____ objects resolves the Y2K38 problem because the upper limits of dates and times are essentially removed.

a. Interval
b. Date
c. DateTime
d. Time

25. To create a DateTime object for the current date, you use the _____ and pass no arguments to the constructor.

a. new keyword
b. time() function
c. date() function
d. new function

26. Type casting refers to PHP code that converts a value from one _____ to another.

a. numeric type
b. data type
c. value
d. character type

27. In an if statement, which is a correct right way to find out whether the string in $a comes before (is less than) the string in $b in a case-insensitive comparison?

a. if ($a < $b)
b. if ((strtolower($a) < (strtolower($b))
c. if (strcomp($a, $b) < 1)
d. if (strnatcomp($a, $b) < 1)

28. With the use of the following escape sequences, what is stored in $message by the code that follows?

```
$message = "The file is in \"C:\\My Documents\"";
```

a. The file is in C:\My Documents Selected
b. The file is in "C:\My Documents"
c. The file is in C:\\My Documents\
d. The file is in "C:\\My Documents\"

29. A list of functions or methods in the reverse order in which they were called is a(n)_____.

a. exception
b. program break
c. method array
d. stack trace

30. If you want to code a catch block that catches all types of exceptions and gets the message associated with the exception, you start the catch block like this:

a. catch (Exception $e) {
b. catch (PDOException $e) {
c. catch (Exception, $e) {
d. catch (PDOException, $e) {

31. Regular expressions

a. can be used to completely validate all types of user entries
b. can only be used to check whether user entries have the right format
c. can be used to completely validate some types of user entries
d. are limited to the validation of string data

32. Runtime errors that occur due to unexpected error conditions are called _____.

a. logical fallacies
b. exceptions
c. breakpoints
d. watchpoints

33. This pattern can be used to validate a five-digit zip code:

a. /[d5]/
b. /+\d[5]?/
c. /^\d{5}$/
d. /$\d5^/

34. To get the message that's related to an exception, you use the

a. message property of the exception object
b. getMessage method of the exception object
c. message property of the exception class
d. getMessage property of the exception class

35. What modifier can be added to the end of a regular expression in PHP to make it case insensitive?

a. i
b. g
c. s
d. a

36. Which character is used to signify the beginning of the string in a PHP regular expression?

a. ^
b. $
c. \b
d. \B

37. Which character is used to signify the beginning or the end of a word in a PHP regular expression?

a. ^
b. $
c. \b
d. \B

38. Which character is used to signify the end of the string in a PHP regular expression?

a. ^
b. $
c. \b
d. \B

39. Which function searches for a regular expression in a string and returns 1 if the pattern is found?

a. match_preg
b. preg_match
c. search_preg
d. preg_search

40. Which of the following are coded patterns that you can use to search for matching patterns in text strings?

a. regular comparisons
b. wild cards
c. regular expressions
d. breakpoints

41. Which of the following will match any character that is **NOT** a letter, number, or underscore in a PHP regular expression?

a. \w
b. \W
c. \d
d. \--

42. Which of the following will match any digit in a PHP regular expression?

a. \w
b. \W
c. \d
d. \9

43. Which of the following will match any letter, number, or underscore in a PHP regular expression?

a. \w
b. \W
c. /d
d. /c